

Practical week-10

Task 1 : Investigate inode structures using commands such as `ls -i`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

```
vennela_07@VennelaPC:~$ mkdir inode_practical
vennela_07@VennelaPC:~$ cd inode_practical
vennela_07@VennelaPC:~/inode_practical$ nano original.txt
vennela_07@VennelaPC:~/inode_practical$ ls -i original.txt
12583 original.txt
vennela_07@VennelaPC:~/inode_practical$ stat original.txt
  File: original.txt
  Size: 78          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 12583         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/vennela_07)   Gid: ( 1000/vennela_07)
Access: 2026-09-26 09:24:20.903532015 +0000
Modify: 2026-09-26 09:24:20.907519041 +0000
Change: 2026-09-26 09:24:20.907519041 +0000
 Birth: 2026-09-26 09:24:20.903532015 +0000
vennela_07@VennelaPC:~/inode_practical$ ln original.txt hardlink.txt
vennela_07@VennelaPC:~/inode_practical$ ls -i
12583 hardlink.txt 12583 original.txt
vennela_07@VennelaPC:~/inode_practical$ stat original.txt
  File: original.txt
  Size: 78          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 12583         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/vennela_07)   Gid: ( 1000/vennela_07)
Access: 2026-09-26 09:24:20.903532015 +0000
Modify: 2026-09-26 09:24:20.907519041 +0000
Change: 2026-09-26 09:24:39.248435900 +0000
 Birth: 2026-09-26 09:24:20.903532015 +0000
vennela_07@VennelaPC:~/inode_practical$ stat hardlink.txt
  File: hardlink.txt
  Size: 78          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 12583         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/vennela_07)   Gid: ( 1000/vennela_07)
Access: 2026-09-26 09:24:20.903532015 +0000
Modify: 2026-09-26 09:24:20.907519041 +0000
Change: 2026-09-26 09:24:39.248435900 +0000
 Birth: 2026-09-26 09:24:20.903532015 +0000
```

```
vennela_07@VennelaPC:~/inode_practical$ cat original.txt
This is the original file.
This file is used to demonstrate inode structures.
This line was added through the hard link.
vennela_07@VennelaPC:~/inode_practical$ ln -s original.txt symlink.txt
vennela_07@VennelaPC:~/inode_practical$ ls -li
total 8
12583 -rw-r--r-- 2 vennela_07 vennela_07 121 Sep 26 09:25 hardlink.txt
12583 -rw-r--r-- 2 vennela_07 vennela_07 121 Sep 26 09:25 original.txt
12556 lrwxrwxrwx 1 vennela_07 vennela_07 12 Sep 26 09:26 symlink.txt -> original.txt
vennela_07@VennelaPC:~/inode_practical$ stat symlink.txt
```

```

vennela_07@VennelaPC:~/inode_practical$ stat symlink.txt
  File: symlink.txt -> original.txt
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 12556         Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/vennela_07)   Gid: ( 1000/vennela_07)
Access: 2026-09-26 09:26:07.614177695 +0000
Modify: 2026-09-26 09:26:02.096862808 +0000
Change: 2026-09-26 09:26:02.096862808 +0000
Birth: 2026-09-26 09:26:02.096862808 +0000
vennela_07@VennelaPC:~/inode_practical$ stat -L symlink.txt
  File: symlink.txt
  Size: 121         Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 12583         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/vennela_07)   Gid: ( 1000/vennela_07)
Access: 2026-09-26 09:25:55.857816354 +0000
Modify: 2026-09-26 09:25:51.003343268 +0000
Change: 2026-09-26 09:25:51.003343268 +0000
Birth: 2026-09-26 09:24:20.903532015 +0000
vennela_07@VennelaPC:~/inode_practical$ stat symlink.txt
  File: symlink.txt -> original.txt
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 12556         Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/vennela_07)   Gid: ( 1000/vennela_07)
Access: 2026-09-26 09:26:07.614177695 +0000
Modify: 2026-09-26 09:26:02.096862808 +0000
Change: 2026-09-26 09:26:02.096862808 +0000
Birth: 2026-09-26 09:26:02.096862808 +0000
vennela_07@VennelaPC:~/inode_practical$ stat -L symlink.txt
  File: symlink.txt
  Size: 121         Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 12583         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/vennela_07)   Gid: ( 1000/vennela_07)
Access: 2026-09-26 09:25:55.857816354 +0000
Modify: 2026-09-26 09:25:51.003343268 +0000
Change: 2026-09-26 09:25:51.003343268 +0000
Birth: 2026-09-26 09:24:20.903532015 +0000

```

```

vennela_07@VennelaPC:~/inode_practical$ ls -li original.txt
12583 original.txt
vennela_07@VennelaPC:~/inode_practical$ find . -inum 123456
vennela_07@VennelaPC:~/inode_practical$ find . -type l -ls
  12556      0 lrwxrwxrwx   1 vennela_07 vennela_07      12 Sep 26 09:26 ./symlink.txt -> original.txt
vennela_07@VennelaPC:~/inode_practical$ readlink symlink.txt
original.txt
vennela_07@VennelaPC:~/inode_practical$ ls -li
total 8
12583 -rw-r--r--  2 vennela_07 vennela_07 121 Sep 26 09:25 hardlink.txt
12583 -rw-r--r--  2 vennela_07 vennela_07 121 Sep 26 09:25 original.txt
12556 lrwxrwxrwx  1 vennela_07 vennela_07  12 Sep 26 09:26 symlink.txt -> original.txt
vennela_07@VennelaPC:~/inode_practical$ rm original.txt

```

```

vennela_07@VennelaPC:~/inode_practical$ cat hardlink.txt
This is the original file.
This file is used to demonstrate inode structures.
This line was added through the hard link.
vennela_07@VennelaPC:~/inode_practical$ cat symlink.txt
cat: symlink.txt: No such file or directory
vennela_07@VennelaPC:~/inode_practical$ original.txt
original.txt: command not found
vennela_07@VennelaPC:~/inode_practical$ stat hardlink.txt
  File: hardlink.txt
  Size: 121         Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 12583         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/vennela_07)   Gid: ( 1000/vennela_07)
Access: 2026-09-26 09:27:46.306542990 +0000
Modify: 2026-09-26 09:25:51.003343268 +0000
Change: 2026-09-26 09:27:15.016420731 +0000
Birth: 2026-09-26 09:24:20.903532015 +0000

```

```
GNU nano 7.2                                     hardlink.txt
This is the original file.
This file is used to demonstrate inode structures.
This line was added through the hard link.
```

Task 2 : Design a program using `mmap()` to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity.

```
vennela_07@VennelaPC:~$ mkdir mmap_demo
cd mmap_demo
vennela_07@VennelaPC:~/mmap_demo$ nano mmap_file.c
vennela_07@VennelaPC:~/mmap_demo$ gcc mmap_file.c -o mmap_file
vennela_07@VennelaPC:~/mmap_demo$ ./mmap_file
Data read using mmap():
Hello! This file is written using mmap().
vennela_07@VennelaPC:~/mmap_demo$ cat mmap_file.txt
Hello! This file is written using mmap().
vennela_07@VennelaPC:~/mmap_demo$ ls -l mmap_file.txt
-rw-r--r-- 1 vennela_07 vennela_07 42 Sep 27 13:06 mmap_file.txt
vennela_07@VennelaPC:~/mmap_demo$ nano traditional_io.c
vennela_07@VennelaPC:~/mmap_demo$ gcc traditional_io.c -o traditional_io
vennela_07@VennelaPC:~/mmap_demo$ ./traditional_io
Data read using read():
Hello! This file uses traditional read/write.
vennela_07@VennelaPC:~/mmap_demo$ ls -l
total 52
-rwxr-xr-x 1 vennela_07 vennela_07 16392 Sep 27 13:06 mmap_file
-rw-r--r-- 1 vennela_07 vennela_07 1062 Sep 27 13:06 mmap_file.
c
-rw-r--r-- 1 vennela_07 vennela_07 42 Sep 27 13:06 mmap_file.
txt
-rw-r--r-- 1 vennela_07 vennela_07 46 Sep 27 13:07 traditiona
l_file.txt
-rwxr-xr-x 1 vennela_07 vennela_07 16360 Sep 27 13:07 traditiona
l_io
-rw-r--r-- 1 vennela_07 vennela_07 611 Sep 27 13:07 traditiona
l_io.c
```

```

GNU nano 7.2                                     mmap_file.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main() {
    const char *filename = "mmap_file.txt";
    const char *data = "Hello! This file is written using mmap().\n";
    int fd = open(filename, O_RDWR | O_CREAT, 0666);

    if (fd == -1) {
        perror("open");
        return 1;
    }

    size_t size = strlen(data);

    if (ftruncate(fd, size) == -1) {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    char *map = mmap(NULL, size, PROT_READ | PROT_WRITE,
                     MAP_SHARED, fd, 0);

    if (map == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    /* Writing to the memory-mapped file */
    memcpy(map, data, size);

    /* Reading from the memory-mapped file */
    printf("Data read using mmap():\n");
    write(STDOUT_FILENO, map, size);

    if (msync(map, size, MS_SYNC) == -1) {
        perror("msync");
    }

    munmap(map, size);
    close(fd);

    return 0;
}

```

```

GNU nano 7.2                                     traditional_io.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    const char *filename = "traditional_file.txt";
    const char *data = "Hello! This file uses traditional read/write.\n";
    char buffer[100];

    int fd = open(filename, O_RDWR | O_CREAT | O_TRUNC, 0666);

    if (fd == -1) {
        perror("open");
        return 1;
    }

    write(fd, data, strlen(data));

    lseek(fd, 0, SEEK_SET);

    int n = read(fd, buffer, sizeof(buffer) - 1);
    buffer[n] = '\0';

    printf("Data read using read():\n");
    printf("%s", buffer);

    close(fd);

    return 0;
}

```